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Combustion dynamics linked to flame behaviour in a partially premixed swirled industrial burner

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Abstract

Previous work [Biagioli, F., Stabilization mechanism of turbulent premixed flames in strongly swirled flows, Combustion, Theory and Modelling 10 (3) (2006) 389–412; Guethe, F., Lachner, R., Schuermans, B., Biagioli, F., Geng, W., Inauen, A., Schenker, S., Bombach, R., Hubschmid, W., Flame imaging on the ALSTOM EV-burner: thermo acoustic pulsations and CFD-validation, in: AIAA Paper 2006-437 presented at the 44th AIAA Aerospace Sciences Meeting and Exhibit, Reno, Nevada, January 9–12, 2006] has shown that turbulent dry low NO_x (partially premixed) flames in high swirl conical burners may be subject to a large change of their anchoring location at the symmetry axis when a critical value of the bulk equivalence ratio is reached, i.e. they are bi-stable. This flame behavior is linked here to combustion pressure dynamics measured in an atmospheric test rig for a prototype version of the Alstom EnVironmental (EV) conical burner. The link is made via the solution of the problem of the 'travelling flameholder', which shows that the unsteady displacement of the flame anchoring location implies an unsteady variation of the flame surface area and therefore unsteady heat release. The relevance of this source of unsteady heat release—which is different from more usual ones due to variations in turbulent burning rate and in the sensible enthalpy jump across the flame—to the generation of combustion dynamics in strongly swirled flows is confirmed here by the strong positive correlation between the tendency of the flame to be displaced and the measured amplitude of pressure pulsations.
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Keywords: Turbulent partially premixed combustion; Large eddy simulation; Combustion dynamics

1. Introduction

The stabilization of turbulent lean premixed flames in industrial burners is usually obtained by the method of vortex breakdown. This consists in the formation of a central recirculation zone (CRZ) which acts as an aerodynamic flameholder, when sufficiently high swirl is given to the flow of reactants. Very often the swirl level is adjusted in a way that vortex breakdown takes place in the downstream free flow (free standing CRZ) without touching any burner wall. Stable anchoring of a turbulent flame via a free standing CRZ zone is a big challenge in the development of low NO_x industrial burners, especially considering that, to keep NO_x emissions below prescribed levels, such anchoring must take places with a good degree of fuel/air premixing.

In the case of an high swirl conical burner with characteristics like the Alstom EnVironmental (EV) double cone burner, it has been shown in [1], via CFD at the level of large eddy simulation (LES) and in [2] via flame imaging techniques, that the properties of the CRZ generated via the vortex breakdown phenomenon have large effect on the stabilization behavior of turbulent premixed flames. This is a direct consequence of the fact that, in strongly non-uniform flows, the position and shape of the flame do not depend only on the mean volumetric consumption rate of reactants for unit of mean flame cross sectional area (the turbulent burning rate U_t) but also on the rate at which reactants are fed into the flame. This point of view is confirmed by the LES analysis performed in [1], which was complemented with a simple one-dimensional boundary

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Nomenclature

c	reaction progress variable
$\mathrm{d}A$	mean flame area element
D_t	turbulent diffusion coefficient
f	fuel mixture fraction
l_t	integral length scale of turbulence
r	radial coordinate
R	radius of combustor
R_F	flame radius of curvature
S	amplitude of flameholder velocity
s_L	laminar flame speed
Sr	Strouhal number
t	time
T_{flame}	adiabatic flame temperature
T_{Ref}	reference flame temperature
U_t	turbulent burning rate
u'	turbulent velocity fluctuation
U	reactants inflow velocity
$\mathbf{u}$	velocity vector
x	axial coordinate
$\dot{x}_F$	flameholder velocity

Greek letters

α	angle between flame and axis
Δp	pressure drop across burner
λ	molecular diffusion coefficient
τ	flame element time lag
ζ	non-dimensional pressure drop across burner
ρ	density
$(\ )_u$	quantity in reactants

layer type model describing flame anchoring at the burner axis. It was shown there that turbulent premixed flames in a swirling flow anchor at the symmetry axis in a position determined by the mean axial convective velocity, the turbulent burning rate U_t and, especially, the turbulent radial inflow of reactants, this last inversely proportional to the flame curvature radius at the symmetry axis R_F. The flame radius of curvature R_F was found to be strongly related to the distribution of the mean axial velocity in the near axis region. it is therefore clear that the position and behavior of the flame are determined also by the characteristics (especially the radial size) of the CRZ and not only by the turbulent burning rate. The results from the analytical investigation were confirmed by experimental data.

A recent experimental investigation [2] has shown additional properties of turbulent premixed flames stabilized in high swirl conical burners. More in details, it was observed that the flame stabilized in these kind of burners are subject to a bi-stable behavior when the flame tip reaches (for example, with a variation in equivalence ratio) a critical position close to the exit of the burner and that a strong correlation exists between this bi-stable flame behavior and the distribution of the measured combustion pressure dynamics.

An hypothesis on the relation between the flame stabilization properties and combustion dynamics in strongly swirled flows is made here with a simple model describing the response of the flame surface area to an unsteady displacement of the flame anchoring location, as it happens close to the critical conditions of flame bi-stability. This mechanism is equivalent to the one proposed in [3,4], where, for the case of a premixed flame anchored at a flameholder fixed in space, the flame surface changes unsteadily due to the time dependent variation of the inflow velocity of reactants. The difficulty in the present case is however represented by the characterization of the flame anchoring properties (which determine the flame anchoring location at the axis) in terms of any unsteady perturbation like, for example, an acoustically modulated equivalence ratio fluctuation. The analytical investigation in [1] has in fact shown that, close to the critical conditions separating the two stable combustion modes in the conical burner, a small variation in equivalence ratio leading to a small displacement of the flame and a small change in the hot gas density can activate a flame–velocity flowfield interaction (very likely of baroclinic nature) which can in turn alter the flame anchoring properties and then further displace the flame. Ideally, the dependency of the flame anchoring location from a perturbation should be expressed on the basis of detailed understanding of the above flame–velocity interaction. Given however the complexity of this problem, an alternative method is proposed here. This consists in determining the flame position-equivalence ratio dependency directly from the experimental or CFD data.

The present work is organized as follows. In Section 2, the behavior of turbulent partially premixed flames in the conical burner is described from an experimental point of view. In Section 3, the flame behavior is studied using CFD and a 1D boundary layer type flame model which was developed in [1]. Finally, in Section 4, the pulsation characteristics of the burner are linked to the flame behavior via a simple model describing the generation of unsteady heat release due to unsteady displacement of the flame anchoring location at the symmetry axis.

2. Experimental investigation of flame behavior in the conical burner

The tests, which are reported in full details in [2], have been performed in an atmospheric single burner combustion test rig at a full-scale industrial burner as described earlier [5,6]. Combustor air, which is electrically preheated and fed into a plenum chamber, flows from there through the burner and the combustion chamber. The combustion

chamber has a rectangular cross-section with the dimensions of 283 mm × 381 mm × 1250 mm. Cooling of the combustor section is achieved by a combination of radiation and water-cooling. The burner geometry is a generic conical one based on the Alstom EV-Burner [7]. This burner shape (see Fig. 1) is obtained by shifting two halves of a cone in opposite directions, orthogonally to the cone axis, such that two inlet slots of constant width are produced. The incoming stream of reactants is therefore given a strong swirl component when flowing through the two inlet slots. Due to the high swirl number, a central recirculation zone, which acts like a flameholder, is formed at the burner axis via the mechanism of vortex breakdown.

An exhaust probe, located about 550 mm downstream of the burner exit, extracts flue gas, which is analysed for contents of oxygen O_2, carbon dioxide CO_2, carbon monoxide CO, unburned hydrocarbons *UHC*, and nitric oxides NO_x (NO and NO_2). The gas analysis data (and the temperatures of combustor inlet air and fuel) as well as the measured mass flows allow a determination of the bulk adiabatic flame temperature T_{flame} in the burner. The accuracy of the absolute T_{flame} measurement is estimated to be approx 25 K. The pressure drop coefficient ζ is obtained from measurement of the pressure difference (Δp) upstream and down stream of the burner non-dimensionalized with the dynamic pressure calculated using a bulk average velocity in the burner. Flames inside the burner having a higher flow resistance lead to higher ζ values. A microphone is placed in the combustion chamber around 180 mm downstream of the burner exit to measure combustion-driven pressure fluctuations. Two air-cooled quartz windows of ca. 200 mm × 200 mm realize optical access to the combustion chamber and are placed 90° to each other and to the mean flow direction. The upstream window edge is located 20 mm downstream of the burner exit in the combustion chamber. To enable a visualization of the pulsations, the test rig is excited with loud speakers mounted upstream of the combustor [8] forcing the system to follow the naturally occurring frequency. The laser set-up [9,10] for the LIF detection of OH radicals and camera were locked on this forcing signal and the phase shift adjusted by manually changing the delay constant. The OH-LIF data were recorded on a regular grid with approx 0.5 mm spacing. The LIF-images are 2D cuts through the combustor along the illuminating laser sheet, which is located in the center of the combustor. The LIF intensity is sensitive to the concentration of OH radicals in their electronic ground states, which are found in regions of heat release, but also in the exhaust gas. Images form chemiluminescence (CL) are also presented.

Fig. 2 shows CL (top) and OH-LIF (bottom) images (total mean, averaging over 300–400 images) in the conical

Fig. 1. Sketch of the Alstom EV burner.

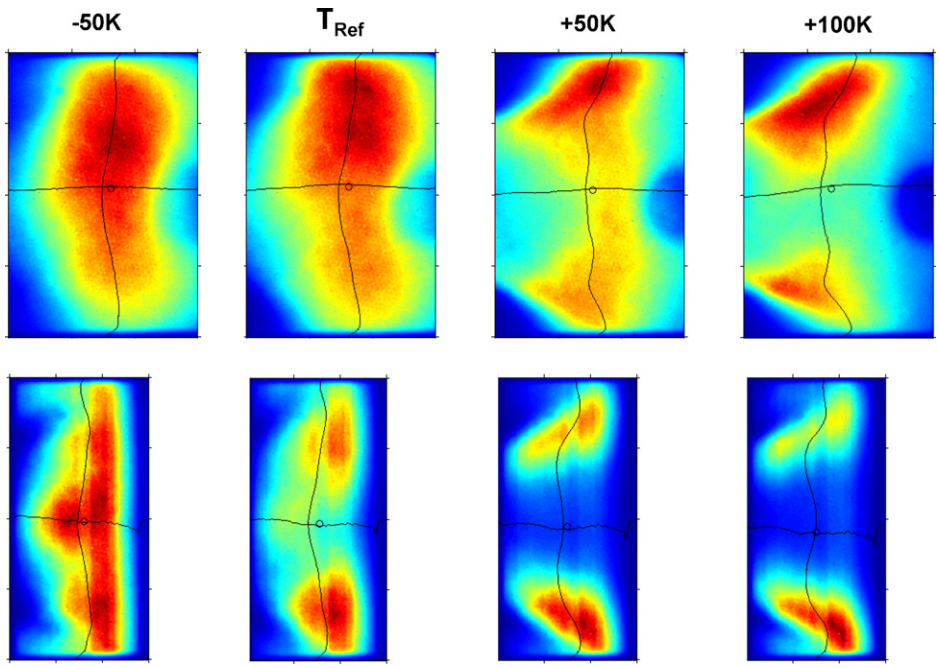

Fig. 2. Flame imaging in the EV burner from chemiluminescence (top) and OH-LIF (bottom).

burner at four different values of the bulk adiabatic flame temperature. The experimental axial velocity flowfield from a water rig facility is instead shown in Fig. 3. The OH-LIF images show that, while the flame at lower adiabatic flame temperatures is located outside of the burner in the combustion chamber, at higher T_{flame} the flame sits rather deep inside the burner. The images indicate that the flame can be stabilized at two different locations: inside or outside of the burner. The transition between the flame 'inside burner' to flame 'outside burner' modes takes place within a rather small variation in flame temperature at the T_{Ref} value,

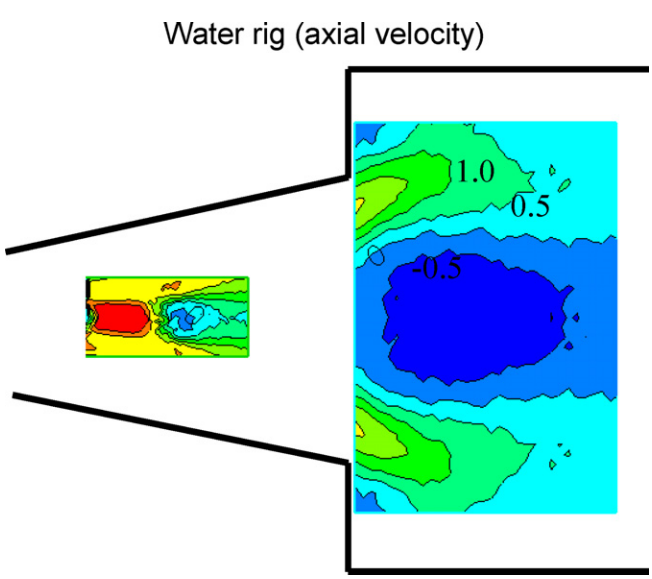

Fig. 3. Contours of mean axial velocity in the conical burner from a water rig facility.

i.e. the flame is bi-stable. With exception of close to the conditions where the transition takes place, both flame positions (inside and outside the burner) seem to be stable over a large range of temperatures. Additionally, the flame anchored inside the burner is not attached to any solid wall of the burner shell, as this would result in a harmful event. It is observed that this flame behavior strongly correlate with the characteristics of the CRZ produced via vortex breakdown. As shown by Fig. 2, in fact, vortex breakdown takes place inside the burner with the appearance of a small CRZ after which low positive axial velocity is recovered and followed by a second CRZ which expands with a large radius in the combustor. Other variants of this burner, which also show a bi-stable flame behavior, show instead a single CRZ that starts very narrow deep into the burner and then grows in thickness in the downstream direction [1].

Interestingly, it is found that the pressure pulsations at the natural acoustic frequency of the combustor (this at a Strouhal number $Sr = 0.57$) have a trend which strongly correlates with the flame behavior. They are in fact rather low at those values of T_{flame} where the flame is stable outside and inside the burner but have a peak at the values of T_{flame} where the transition between the two flame modes takes place. This is shown in Fig. 4 (left), where also the pressure drop in the burner is plotted which, increasing with the displacement of the flame anchoring location inside the burner, is an excellent indicator of flame position (see the following section). The strong link between the pressure pulsations and the flame behavior becomes clear from images at $T_{\text{flame}} = T_{\text{Ref}}$ phase locked to the forcing signal (see Fig. 5), where the pressure pulsations amplitude versus mean flame temperature attains the maximum value.

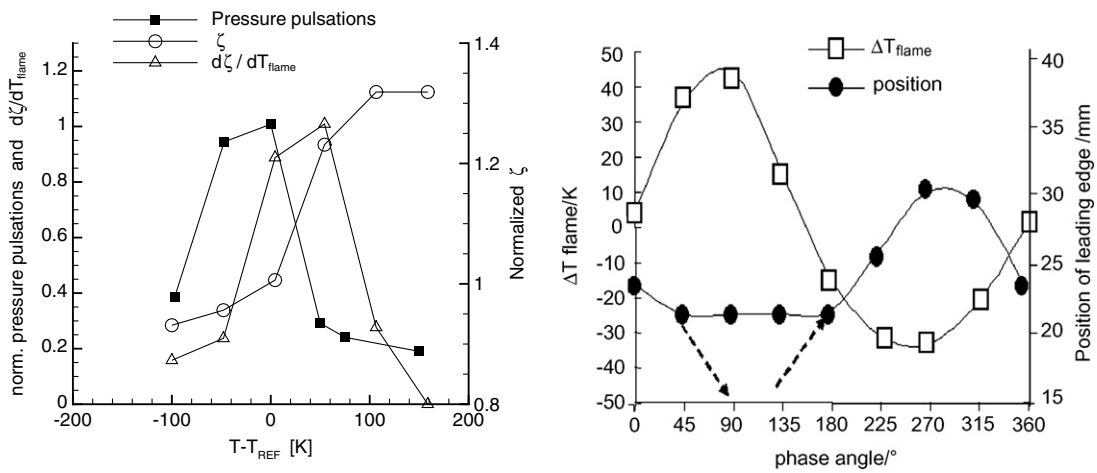

Fig. 4. Left: burner non-dimensional pressure drop ζ, $d\zeta/dT_{flame}$ and amplitude of combustion dynamics versus bulk flame temperature. Right: mean phase locked position of the flame tip and flame temperature from chemiluminescence plotted versus the flame angle from the case at the bulk adiabatic flame temperature $T = T_{Ref}$.

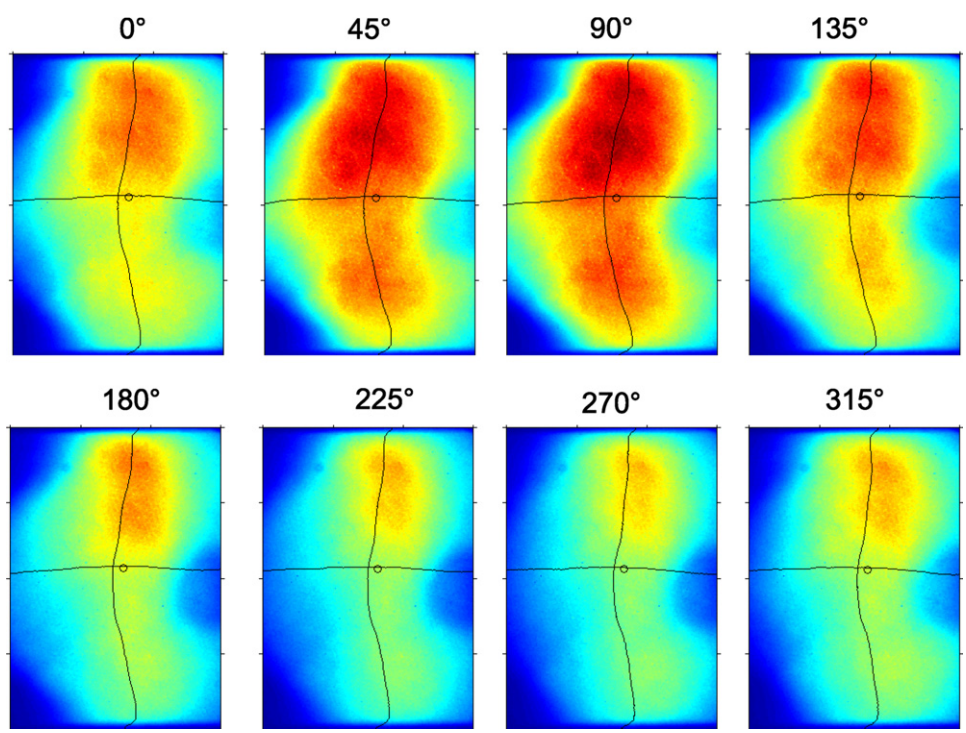

Fig. 5. Flame imaging in the EV burner from chemiluminescence. Phase locked averging.

Resolving the phase angle shows that the flame is moving between the two positions where the flame can be stable, similarly to the effect on the total mean of variations in bulk T_{flame}. This behavior is resumed in Fig. 4 (right), reporting phase locked values of T_{flame} (these estimated from chemiluminescence intensity and due to fluctuations in equivalence ratio induced by the external forcing of the air mass flow rate) and the position of the leading edge of the flame as function of the phase angle. The dependency of the flame temperature T_{flame} from the chemiluminescence intensity I_{CL} is calibrated using the totally averaged chemiluminescence data.

3. Analysis of vortex breakdown in industrial burners via Large Eddy Simulation

The problem of vortex breakdown in the conical burner is investigated here with the help of CFD. Because Reynolds Averaged Navier–Stokes (RANS, k–ϵ approach) methods have been previously [1] found unable to correctly describe the kinematic properties of this kind of flows (with strong impact on the accuracy with which the stabilization of turbulent premixed flames could be predicted), the present investigation is based on large eddy simulation (LES) performed using the commercial solver FLUENT

version 6.1.22. All geometric quantities and velocities are non-dimensionalized with the burner diameter D and burner bulk velocity U_0. The combustor has circular cross-section with non-dimensional diameter and length, respectively, equal to 2 and 7. The computational domain has been meshed with a tetrahedrals mesh of approx. 1.3 million. The finite volumes elements inside the burner and in the combustor close to the burner exit have an average non-dimensional size of 0.025. Turbulent transport is modelled with the standard Smagorinsky method. Convective terms are discretized with the central differencing scheme for all equation with exception of the fuel mixture fraction equation which is discretized with a second-order upwind scheme. The partially premixed turbulent premixed flame is simulated with the Turbulent Flame Speed model from Zimont [11]. This model represents the flame as a propagating boundary layer which increases in thickness according to the turbulent dispersion law and propagates with speed given by the expression:

$$U_t = Au'^{0.75} l_t^{0.25} \lambda^{-0.25} s_L^{0.5} \tag{1}$$

where the laminar flame speed s_L is calculated with the premix module of the CHEMKIN II package using GRI3.0 [12] chemical kinetics at the local value of the mean mixture fraction $\tilde{f}$. The mean progress variable transport equation of the TFC model is therefore given by

$$\frac{\partial(\overline{\rho}\tilde{c})}{\partial t} + \nabla \cdot (\overline{\rho}\tilde{\mathbf{u}}\tilde{c}) = \nabla \cdot (\overline{\rho} D_t \nabla \tilde{c}) + \rho_u U_t |\nabla \tilde{c}| \tag{2}$$

The main outcome of these CFD–LES simulations is in line with the results obtained in [1] for a variant of the EV burner with higher swirl number. Until a given critical value of the adiabatic flame temperature, the turbulent premixed flame stabilizes outside the burner with its tip very close to the burner exit (see Fig. 6, left). Interestingly, the flame completely intersects the CRZ indicated via the white $\tilde{u} = 0$ contour. It should be observed that the structure of the CRZ in front of the flame is in line with the one measured in the water rig facility (see Fig. 4, left), i.e. starting small and narrow deep inside the burner, then contracting and expanding again more downstream. When this critical value of the adiabatic flame temperature is just slightly exceeded, the flame anchoring location is subject to a large displacement into the burner (see Fig. 6, right) at the apex of the CRZ. For higher values of the adiabatic flame temperature, the flame anchoring position does not change significantly anymore. Fig. 7 (left) shows also that, at the critical conditions of the flame displacement into the burner, the pressure drop across burner and flame has a

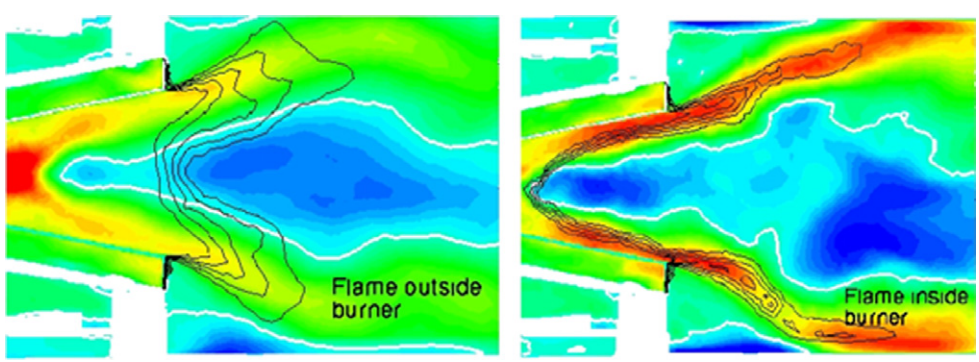

Fig. 6. Results from LES analysis. Left: mean progress variable isolines and velocity contours at $T < T_{Ref}$. Right: at $T > T_{Ref}$.

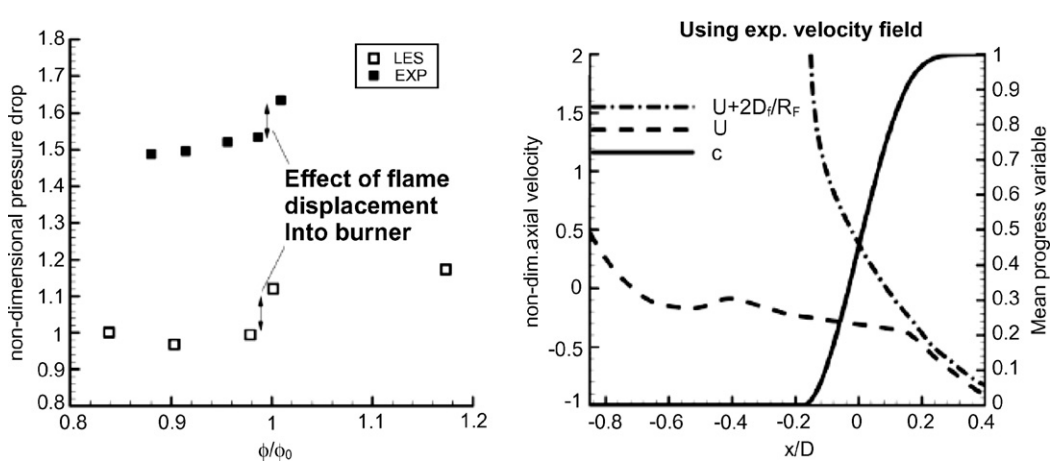

Fig. 7. Left: comparison of the pressure drop across the flame from LES with experimental data. Right: estimation of the mean progress variable distribution at the axis from 1D boundary layer method developed in [1] using the velocity distribution measured in a water rig.

discontinuity in line with the one measured in the experiments (note that this figure refers to a variant of the burner which, due to both the higher swirl number and fully premixed conditions, is characterized by a more sudden transition between the two flame stabilization modes). Such bi-stable flame behavior might be interpreted as an obvious transfer of the flame anchoring location between two consecutive CRZs at the axis separated by a contraction which represent a recovery of positive axial velocity. The investigation performed in [1] shows however that the situation is much more complex than just a jump of the flame from one CRZ to the other. In [1], a boundary layer method was developed to deal the problem of flame anchoring at the symmetry axis in strongly swirled flows. It was assumed that the velocity flowfield approaching the flame is like the one measured in a water rig facility (non-reacting velocity flowfield). Using a second-order Taylor expansion in the radial direction, it was shown that the 2D mean progress variable transport equation can be written as the following 1D equation (valid along the symmetry axis):

$$\overline{\rho}\left(\tilde{u}+\frac{2}{R_{\mathrm{F}}}D_{\mathrm{t}}\right)\frac{\mathrm{d}\tilde{c}}{\mathrm{d}x}=\frac{\mathrm{d}}{\mathrm{d}x}\left[\overline{\rho}D_{\mathrm{t}}\frac{\mathrm{d}\tilde{c}}{\mathrm{d}}\right]+\rho_u U_{\mathrm{t}}\left|\frac{\mathrm{d}\tilde{c}}{\mathrm{d}x}\right| \tag{3}$$

This equation shows that the radial turbulent transport term $\simeq D_{\mathrm{t}}\partial^2\tilde{c}/\partial^2 r$ in the 2D progress variable equation written in a cylindrical coordinate system is transformed at the axis in the term $2(D_{\mathrm{t}}/R_{\mathrm{F}})\mathrm{d}\tilde{c}/\mathrm{d}x$ which has a convective form and where R_{F} is the flame radius of curvature at the axis. The radius R_{F} of the flame at the symmetry axis is obtained from a second transport equation (not reported here) for the second-order contribution in the Taylor expansion of the progress variable $\tilde{c}$ in radial direction. In first approximation the flame radius of curvature is inversely proportional to the radial variation of the mean axial velocity, i.e. the factor $\partial^2\tilde{u}/\partial r^2$ calculated at the symmetry axis.

The application of this method using the velocity flowfield from the water rig shows that the flame cannot enter into the burner because there the inflow of reactants into the flame $\tilde{u}+2D_{\mathrm{t}}/R_{\mathrm{F}}$ would be much higher than how much the flame can burn, i.e. than the turbulent burning rate U_{t}. This result is given in Fig. 7 (right) where the calculated mean progress variable, the terms $\tilde{u}$ and $\tilde{u}+2D_{\mathrm{t}}/R_{\mathrm{F}}$ along the symmetry axis are plotted. It must be observed that, the flame would be accommodated deep into the burner if the turbulent burning rate was balanced only with the mean axial inflow of reactants ($\tilde{\rho}\tilde{u}=\rho_u U_{\mathrm{t}}$). Inside the burner however R_{F} would be very small, due to very narrow CRZ, which is the reason of the high radial turbulent inflow $2D_{\mathrm{t}}/R_{\mathrm{F}}$. For this reason, if the velocity flowfield is kept like the non-reacting one, the flame does not move inside the burner even with a very high increase in the turbulent burning rate U_{t}. This strong resistance of the baseline non-reacting velocity flowfield to the flame moving inside the burner was confirmed in [1] with LES simulations where the density ratio across the flame was forced to be constant. This indicates that, in the variable density simulations where the flame moves rather suddenly into the burner (see Fig. 6 and the sudden change in the pressure drop across the burner in Fig. 7, left) at a critical value of the bulk adiabatic flame temperature, a flame–velocity interaction must take place which acts on the structure of the approaching velocity flowfield (for example, changing $\partial^2\tilde{u}/\partial r^2$ at the axis). Even if a full study has not been yet performed, the indication, also from other investigators [13], is that such an interaction consists in the generation of baroclinic azimuthal vorticity due to the non-alignment of density gradient across the flame and pressure gradient in the radial direction due to flow centrifugation. This added vorticity of baroclinic nature has the effect of inducing (according to the Biot–Savart law, see [14]) negative axial velocity in the near axis region.

4. Relation between unsteady displacement of the flame anchoring point and combustion pressure dynamics

It has been shown in the previous section that turbulent premixed flames stabilized in the high swirling flow produced by a conical burner are bi-stable. Below a critical value of the bulk equivalence ratio in the burner, the flame is stabilized in the combustor with its tip close to the exit of the burner and above this critical value the flame is anchored with its tip deep into the burner. It is observed, via phase averaging of chemiluminescence data, that, at the critical conditions, the flame anchoring location is characterized by large sensitivity to equivalence ratio fluctuations. The experiments also show that, with increasing mean equivalence ratio in the burner, the amplitude of combustion dynamics first increases, reaching a peak slightly before the critical conditions for rapid flame accelaration, then decreases, when the flame has steadily anchored inside the burner. Finally, it has been shown via LES and a simple 1D approach, that the strong sensitivity of the flame anchoring location to perturbation in equivalence ratio at the critical conditions cannot be explained just in terms of the change in turbulent burning rate but it must necessarily imply a rather complex interaction between flame and aerodynamics of the burner which leads to a modification of the flow in the burner and hence of the local flame anchoring properties.

From a thermo-acoustics point of view, fluctuations in acoustic pressure are generated by fluctuations in heat release, more directly by fluctuations in the volumetric rate of reactants consumption (or, equivalently, in products formation). The total volumetric rate of reactants consumption is given by $\dot{V}_R=\int_A U_{\mathrm{t}}\,\mathrm{d}A$ where U_{t} is the turbulent burning rate, i.e. the volumetric rate of reactants consumption for unit of mean flame cross-sectional area and $\mathrm{d}A$ an infinitesimal element of mean flame area. The rate of heat release associated to $\dot{V}_R$ is given by $\dot{Q}=\int_A \rho_u f H_{\mathrm{F}} U_{\mathrm{t}}\,\mathrm{d}A$, where H_{F} is the fuel calorific value, f the fuel mass fraction, ρ_u the local density of reactants. This relation indicates that unsteady heat release can be

generated by fluctuations at the flame in: reactant density ρ_u, fuel mixture fraction f (or equivalently the equivalence ratio ϕ), turbulent burning rate U_t and mean flame surface area A. Fluctuations in U_t are usually connected to fluctuations in the parameters controlling this quantity, i.e. the turbulent velocity fluctuation u' and the fuel mixture fraction f at the flame. In order to determine the stability of the burner–flame–combustor system, fluctuations in mixture fraction f at the flame (which also directly affects the heat release rate) are put in relation, via a convective time lag, with the f fluctuations generated at the position of fuel and air injection by variations in acoustic velocity. The generation of unsteady heat release based only on these mechanisms seems however not sufficient to explain the combustion dynamics levels observed in the present experiments. Assuming in fact that acoustic forcing in the present experimental tests has the same amplitude at all test conditions and assuming that there is not large amplification from the system (due to a combustion instability issue), also fluctuations in equivalence ratio, hence in U_t, $\dot{Q}$ and the combustion dynamics amplitude should be constant.

The problem mentioned above can be solved if the contribution to the unsteady heat release originating from fluctuations in the mean flame surface area A is also accounted for. These fluctuations in A arise from the unsteady displacement of the flame anchoring location due to a perturbation, displacement which becomes strong when the flame reaches a critical position at the burner axis. This problem is investigated here in a qualitative way, considering the flame flat and infinitesimally thin, anchored at the axis in a uniform flow, with an angle to the axis determined by the mean flow velocity U and the turbulent burning rate U_t. It is assumed that the flame anchoring location is subject to an unsteady displacement with velocity $\dot{x}_F(t)$ (negative in the downstream direction) whose dependency from equivalence ratio fluctuations will be clarified later. The turbulent burning rate U_t is assumed constant. When the flame moves unsteadily between two positions, unsteady heat release is generated because of the consumption of the reactants volume included between the two positions. We study this problem of the "travelling flameholder" in the frame of reference travelling with the flame anchoring location. In this frame of reference the effect becomes equivalent to a variation in the velocity of the incoming reactants into $u(t) = U + \dot{x}_F(t)$, i.e. equivalent to the problem addressed in [3] via the use of the so-called G equation which tracks a propagating front of infinitesimal thickness. It is assumed that, at the initial time $t = t_1$, until which the flame has been steady with an angle to the axis given by mass conservation as $\alpha_0 = \arcsin(U_t/U)$, the flame anchoring point changes its speed from zero to $\dot{x}_{F,1}$ in the upstream direction as shown in Fig. 8, top. The flame starts reacting from its anchoring location to this perturbation as follows. In the frame of reference moving with the flame anchoring point, the convective velocity of reactants through the flame becomes $U + \dot{x}_{F,1}$ such that, starting from $t = t_1$, the flame leaves the anchoring point with an angle $\alpha_1 = \arcsin[U_t/(U + \dot{x}_{F,1})]$ which is smaller than the initial angle $\alpha_0 = \arcsin(U_t/U)$. Those elements of the flame which have been released at $t < t_1$ are instead only convected downstream at the new velocity which implies that they keep the initial angle α_0 to the axis. At $t = t_2$ the flame anchoring location changes again its speed to $\dot{x}_{F,2}$ in the downstream direction and the new shape assumed by the flame can be estimated also as explained above. The shape of the flame changes, at different time steps, as shown in Fig. 8 (top). The bottom part of the figure shows instead the response of the flame to an harmonic variation in the velocity of the anchoring location (calculated according to the above procedure as a

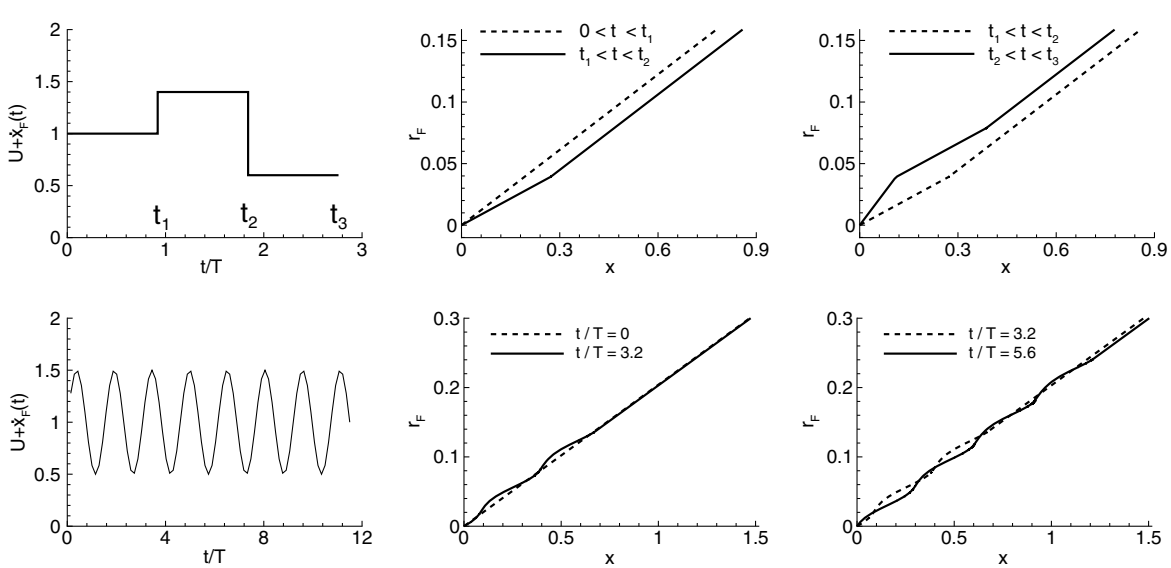

Fig. 8. Evolution of flame area with displacement of the flame anchoring location. Top: due to a stepwise variation in the velocity of the flame anchoring location at the axis. Bottom: due to harmonic variation. Time non-dimensionalized with the bulk residence time in the burner.

sequence of many small steps whose amplitude is distributed harmonically in time).

For a more generic time variation of the flameholder velocity $\dot{x}_F(t)$, we proceed as follows. According to the previous discussion, the angle of the flame element at a given radial location r and time t can be calculated as the angle which the flame elements had at the time $t-\tau(r)$, when it was released by the flameholder:

$$\sin\alpha(r,t) = \frac{U_t}{U + \dot{x}_F[t-\tau(r)]} \tag{4}$$

where $\tau(r)$ is the time delay at which the flame element at radius r was released by the flameholder. Therefore the angle of the flame at radius r is the angle of the flame at the flame anchoring location calculated with constant U_t and the inflow velocity at the time the flame element was released. Assuming small perturbations, we estimate here this time using the mean radial velocity of the flame element $U_t\cos\alpha_0$.

$$\tau(r) = \frac{r}{U_t\cos\alpha_0} \tag{5}$$

the area of the flame element at radius r and time t is therefore given by

$$dA(r,t) = \frac{2\pi r dr}{\sin[\alpha(r,t)]} = \frac{2\pi r\{U + \dot{x}_F[t-\tau(r)]\}dr}{U_t} \tag{6}$$

assuming a periodic variation in the velocity of the flame anchoring location we have (for $t > \tau(R)$ with R maximum radius):

$$\dot{x}_F[t-\tau(r)] = S\sin[\omega(t-r/U_t)/\cos\alpha_0] \tag{7}$$

where H is the step function to account for the perturbation starting at $t = t_1$. The integration of this equation between 0 and the maximum radius R yields

$$A(t) = \pi R^2\frac{U}{U_t} + \frac{2S\cos\alpha_0}{\omega}\left\{R\cos\omega\left(t - \frac{R}{U_t\cos\alpha_0}\right) + \frac{1}{\omega}U_t\cos\alpha_0\left[\sin\omega\left(t-\frac{R}{U_t\cos\alpha_0}\right) - \sin\omega t\right]\right\} \tag{8}$$

where the first term represent the mean flame area and the second one the fluctuation in mean area due to the unsteady displacement of the flame anchoring location. In order to close the problem, the amplitude S of the fluctuation in flame anchoring location must be expressed in terms of the controlling parameters. As already discussed, the relation between the fluctuation in equivalence ratio and the position of the flame anchoring location is a complex problem which involves a not yet fully understood flame–velocity interaction mechanism. The dependency of the flame anchoring location from the equivalence ratio (or, equivalently, the adiabatic bulk flame temperature T_{flame} in the burner) can be however worked out directly from experiments or from CFD results. For example, at a qualitative level, it can be assumed that the flame position is proportional to the pressure drop coefficient ($x_F = k\zeta$) and then obtain an index of the sensitivity of the flame anchoring position at the axis to T_{flame} as $dx_F/dT_{flame} = k d\zeta/dT_{flame}$. This quantity, which is shown in Fig. 4, reaches a peak close to the T_{flame} where combustion dynamics also attain their peak confirming the idea presented here. We assume here that the flame position responds dynamically to the variations in equivalence ratio following the steady state characteristic behaviour (quasi-steady-state assumption). In reality, it can be expected that the mechanism of flame–aerodynamics interaction which is responsible for altering the flame anchoring conditions and then displacing the flame will become effective after a characteristic time delay to the incoming equivalence ratio perturbation. It follows therefore that the effectiveness of such a mechanism decreases with increasing frequency. Given the quasi-steady state assumption on the flame response, the fluctuation in flame position due to a fluctuation in flame temperature around a mean value $T_{flame\star}$ can be expressed as

$$x'_F = k\left.\frac{d\zeta}{dT_{flame}}\right|_{T_{flame\star}} T'_{flame} \tag{9}$$

The speed $\dot{x}_F$ at which the flame anchoring location moves can be worked out from this relation for an equivalence ratio fluctuation with a given frequency $f = \omega/2\pi$. The modulation of the equivalence ratio at the flame could be, for example, due to the change in acoustic velocity (hence in air mass flow rate) at the location where fuel is injected. Therefore, given the harmonic fluctuation in adiabatic flame temperature:

$$T'_{flame} = -\Delta T_{flame}\cos(\omega t) \tag{10}$$

we obtain from (7), (9) and (10):

$$S = k\left.\frac{d\zeta}{dT_{flame}}\omega\right|_{T_{flame\star}} \text{‘}\Delta T_{flame} \tag{11}$$

Introducing this expression into (8) yields the transfer function between the fluctuating flame surface area and the fluctuations of adiabatic flame temperature in the burner. This is a fundamental ingredient in the network based analysis [15] of the thermo-acoustics stability of combustor devices. This contribution has to be added to the classic one given by variations in turbulent burning rate.

5. Conclusions

It has been shown that flames in high swirled flows undergoing vortex breakdown are characterized by complex stabilization properties. The analysis has been carried out in the case of a conical burner with characteristics like the Alstom EV one. It is shown that the narrowing of the CRZ inside the burner is responsible for bi-stable behavior of the flame, very likely driven by flame–velocity flowfield interaction. Close to the critical conditions separating the two stable positions of the flame (inside and outside the burner), the flame anchoring location is strongly sensitive

to flow and equivalence ratio perturbation. This generates unsteady heat release via the unsteady creation and destruction of the flame surface area which accounts for the consumption of the reactants included between the two stable positions. It is shown that combustion pressure dynamics produced via acoustic forcing of the burner at its natural acoustic frequency have a very strong relation to the unsteady heat release generated in this way. Future work should clarify the nature of the flame–velocity interaction at the critical conditions and also address the role of the present source of unsteady heat release within the problem of combustion instability which has not been considered here.

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